

ELI5 how is Ai bad for the environment

Posted by u/jitted_timmy • 0 points • 44 comments

I have a vague understanding of how technology in general is harmful to the environment, obviously it consumes energy, but how does Ai do significantly more damage? How is wated involved/impacted?

Comments

u/Fit-Percentage-6075 • 1 points • 2025-10-15 19:37 UTC

Ai isn't bad for the environment, ai doesn't live, breathe, think, create technology, destroy forests, start wars, let greed consume it...if you really want to point a finger at something point it at humanity itself. We are the vampire and the world our victim. Either way all things come to an end eventually. Overpopulation is the biggest issue of the world right now as humans have no natural predators and we kill everything that looks at us wrong. Either way the suns gonna explode at some point and none of it will matter anyways. Live your life to the fullest you only get one.

u/jitted_timmy • 1 points • 2025-10-15 22:34 UTC

This did not answer my question lol

u/[deleted] • 1 points • 2025-08-14 18:43 UTC

[removed]

u/Miliean • 1 points • 2025-05-22 19:29 UTC

> obviously it consumes energy

It's really just this. It's important to know that because of how our current AI models work, it takes A HELL of a lot of computer horsepower and time just to run simple queries. All that horsepower, it comes from electricity. There's really no magic here, they have very powerful computer servers and those very powerful servers drink up A LOT of electricity.

Exact numbers are VERY hard to come by, but in general every single query in ChatGPT cost something like 10-100x what it costs to run a google query. And mostly that's the cost of the electricity to run the computers both for the query itself but also to train the AI model in order to be able to answer the query in the first palce.

u/Envoyager • 0 points • 2025-05-22 02:48 UTC

Think about it this way. Relatively speaking, AI, like chatgpt, is still a child. Children are chaotic, running around, burning more calories. Children also run a warmer temperature than an adult for physiological reasons. As they grow older they start to slow down and cool off, all the while getting smarter and more efficient. Over time, AI will have consumed and learned all it can and will need less electricity and run

cooler. AI engineers and scientists will continue working on new ways to make the data centers more efficient as well

u/Whiskeytangr • 0 points • 2025-05-22 02:15 UTC

In a very broad sense, from a particular school of Anthropology, there are various epochs in how humans have organized themselves and the available resources. These are called modes of production, and are accompanied by a technological innovation supporting the resource/population need. Need first, technology follows the need. Hunter/gatherer -> pastoral/horticultural -> agricultural -> industrial -> service -> information -> ???

What is not often discussed is each of those progressions represent a decrease in overall efficiency, but can accommodate a higher total output, to orders of magnitude. No real #'s here but for example the progression is 1:1 calorie expense for gatherers, 1:1million calorie expense for industrial cultures. Point being, efficiency goes down, overall production goes up as it responds to population needs.

Where this inefficiency equals insanity is that you cannot continually throw more energy input into the system exponentially. Google a blog called Do The Math. Surface temp of the planet will be billions of degrees if you just keep adding energy, doesn't matter if it's coal, solar, fusion, or other.

Tldr;

AI can be assumed to (and does) take a lot more energy than the previous mode of production to produce the same amount. Is the increased production needed to sustain the current and projected population, y/n? If yes, can the earth sustain the exponential growth, or does resource allocation have to extend to beyond? Yes AI is bad for the environment, there is zero way around that.

u/dressthr0w • 1 points • 2025-05-22 02:11 UTC

A lot of replies point out to how much energy AI [uses](<https://calcubest.com/articles/AI-energy-use/>). But something else to keep in mind is that so much use of AI is mundane tasks that a simple google search would suffice. Things like looking up how many cups in a gallon, how old the president is, etc. Things like chatbots, or customer support bots or whatever.

AI proponents point to the possible use of AI in helping with medicine and research and making works more productive. All of these are amazing goals. But I have a feeling most AI use is actually stuff that isn't nearly as important, and so the benefits of using AI might not be worth the costs (like the huge amounts of energy used).

Tech companies love to promote things are being revolutionary and benefiting society in amazing ways, and in reality we get products that are a lot more lackluster.

u/ZeusThunder369 • 1 points • 2025-05-22 02:05 UTC

What isn't commonly known is that "AI" (usually referring to an LLM, but sometimes machine learning) is built using brute force. It isn't elegant at all.

The GPUs that power modern AI are very power hungry at scale. That power usage is bad for the environment.

u/Kingreaper • 0 points • 2025-05-22 01:50 UTC

AI uses about 0.42% of the world's electricity.

People who hate AI like to compare the entirety of all the energy used by AI to something like a single home computer. That makes it seem HUGE. But that's not a fair comparison.

It's 14% - 1/7th - of the total energy usage of big server farms - and requires no more water than the other big server farms.

Is that a lot? Is 1/200th of the world's power consumption too much? That's for you to judge.

Or we can compare it on the scale of single uses. A ChatGPT prompting uses about 10x as much electricity as a google search. Is that too much?

Getting dozens of prompted images from an AI costs less electricity than having a digital artist draw one on their PC. Is that too much?

u/Jaymac720 • 1 points • 2025-05-22 01:49 UTC

Computer servers use a lot of energy. With the high power required for AI programs, they suck down a ton of energy. They also need cooling which requires big heat pumps or water submersion